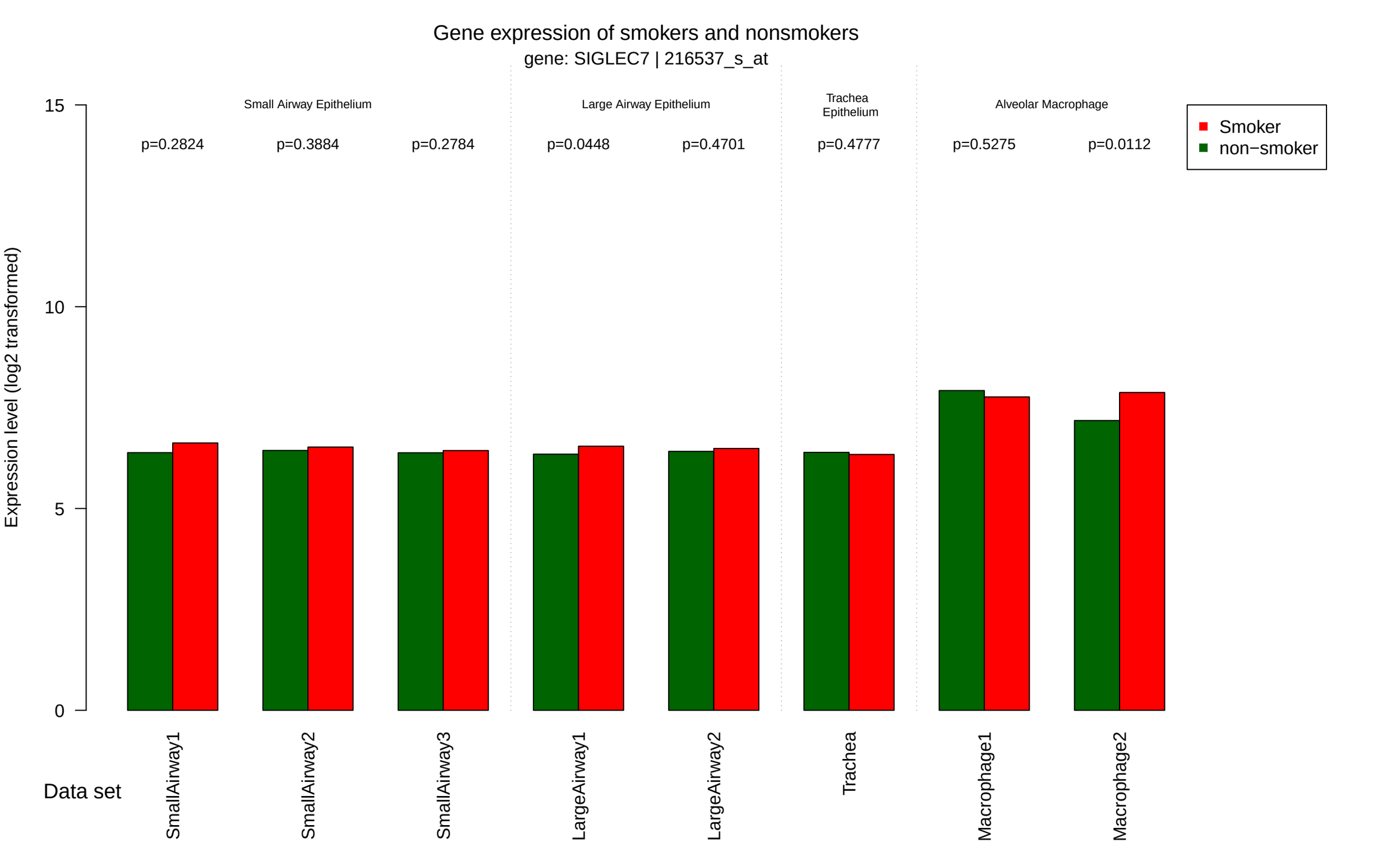
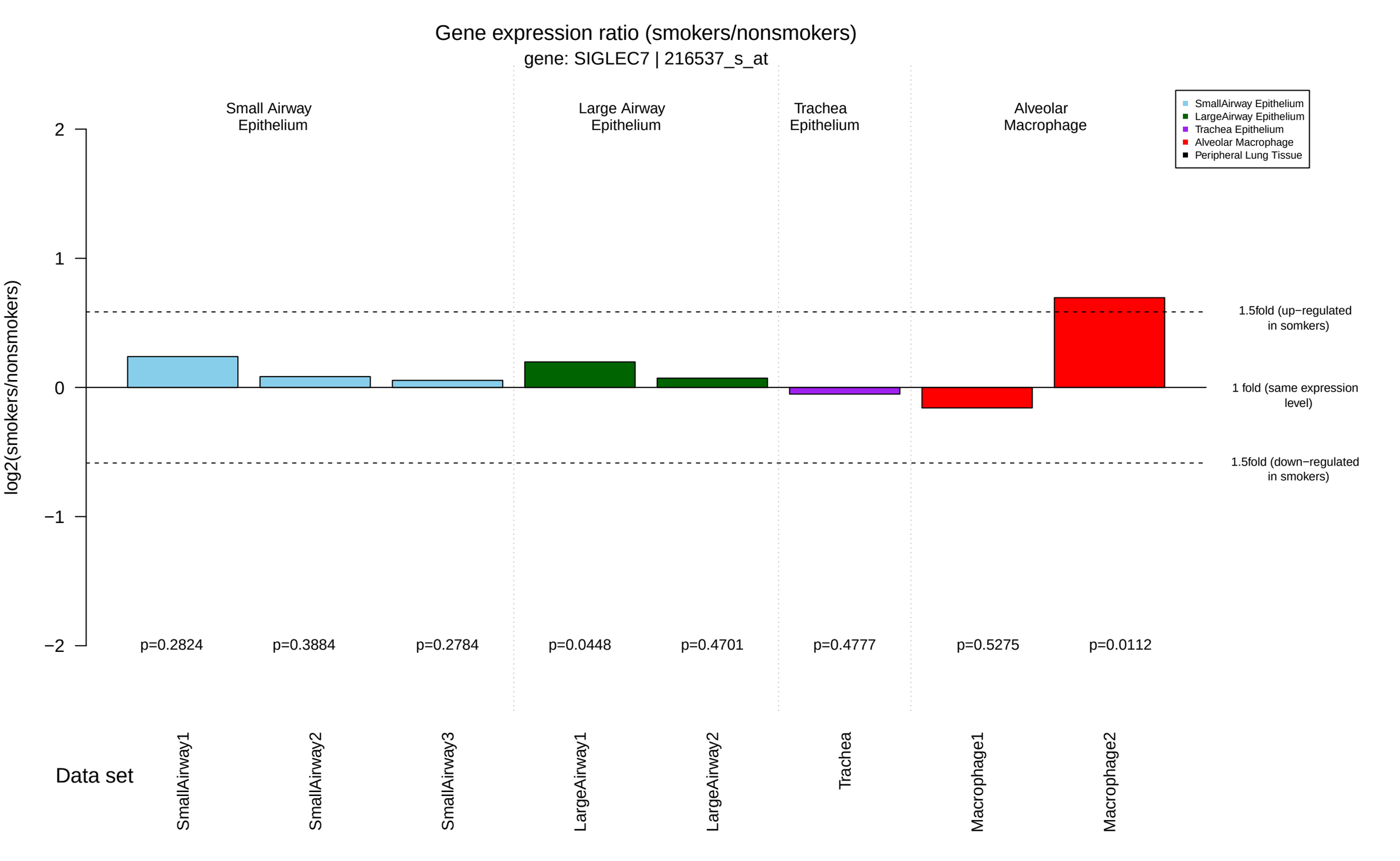
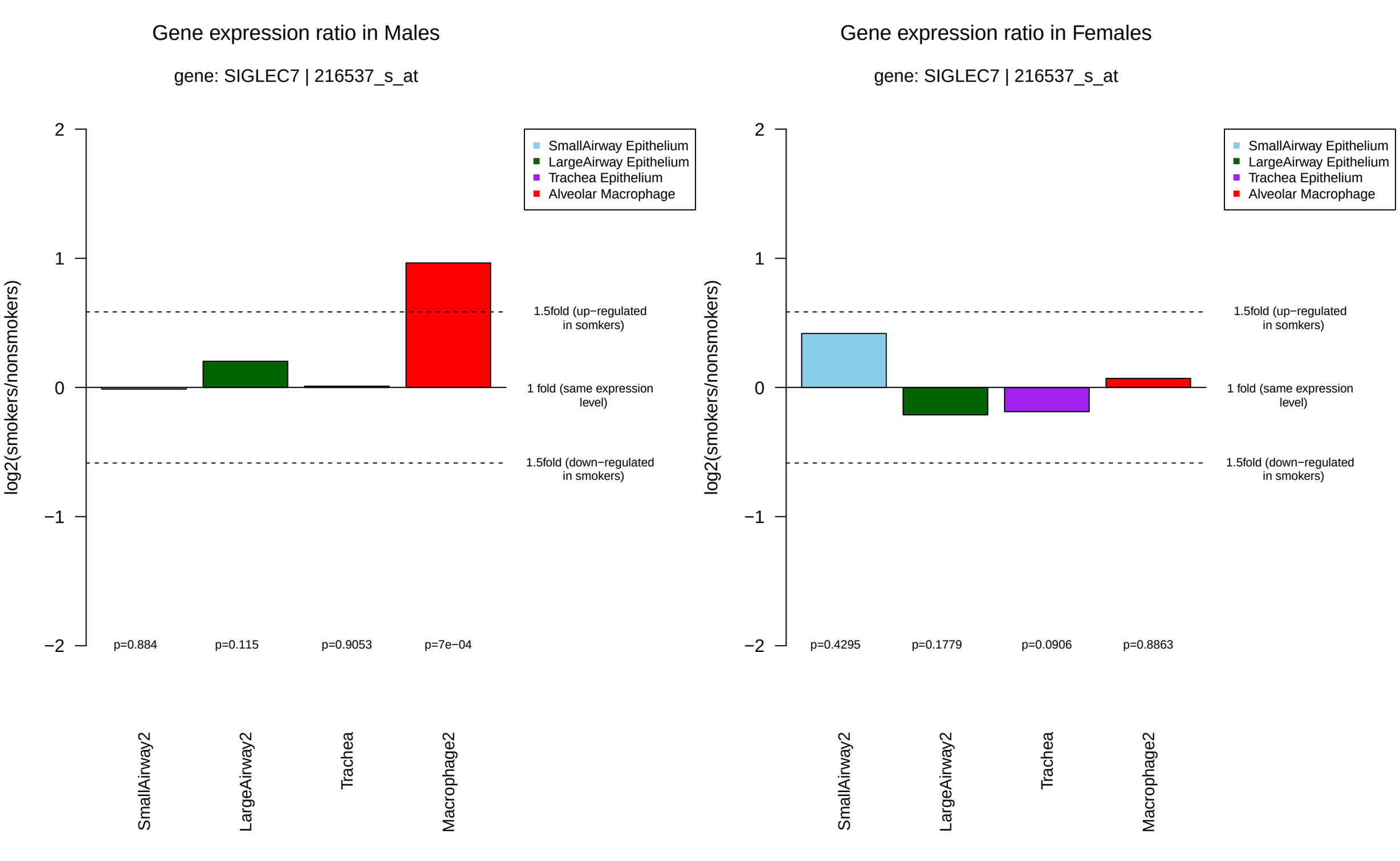
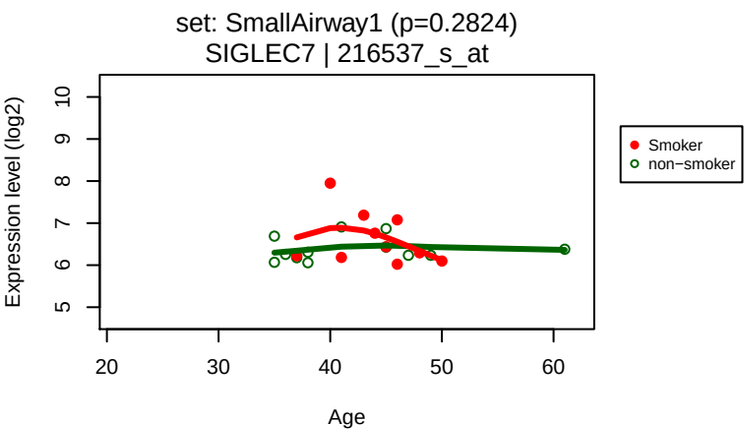
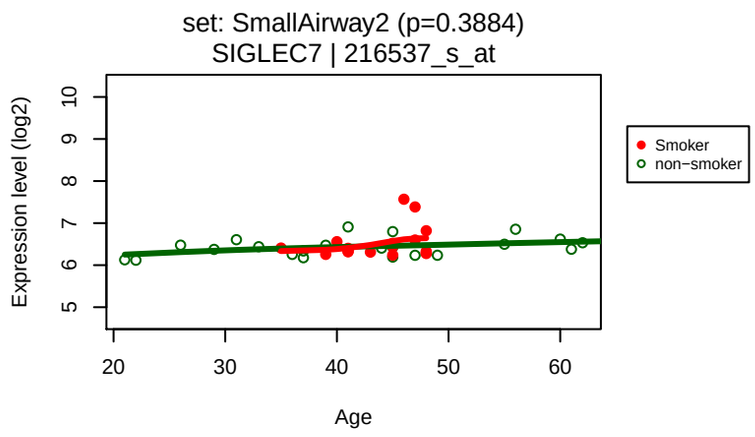
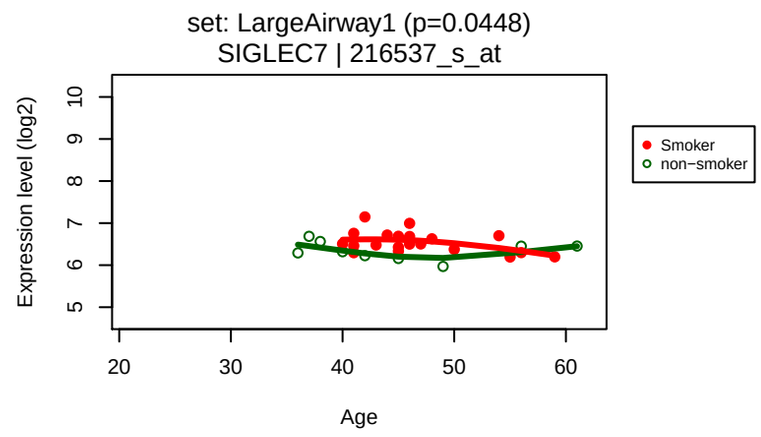

Gene expression of smokers and nonsmokers
gene: SIGLEC7 | 216537_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2824
p=0.3884
p=0.2784
p=0.0448
p=0.4701
p=0.4777
p=0.5275
p=0.0112
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

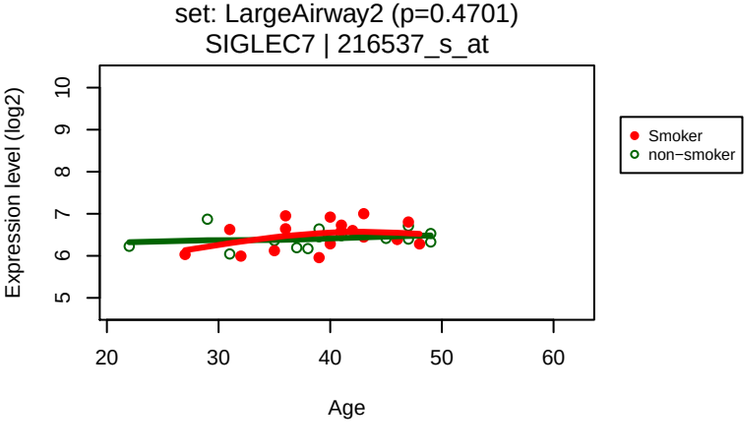

Gene expression ratio (smokers/nonsmokers)
gene: SIGLEC7 | 216537_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2824
p=0.3884
p=0.2784
p=0.0448
p=0.4701
p=0.4777
p=0.5275
p=0.0112
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

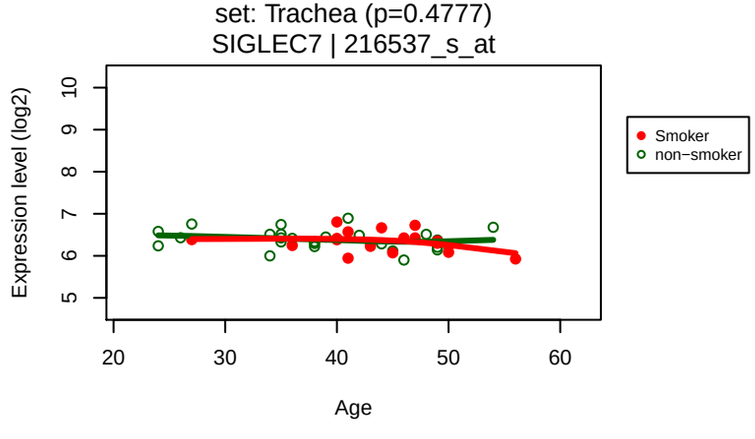

Gene expression ratio in Males
gene: SIGLEC7 | 216537_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.884
p=0.115
p=0.9053
p=7e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIGLEC7 | 216537_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4295
p=0.1779
p=0.0906
p=0.8863
SmallAirway2
LargeAirway2
Trachea
Macrophage2

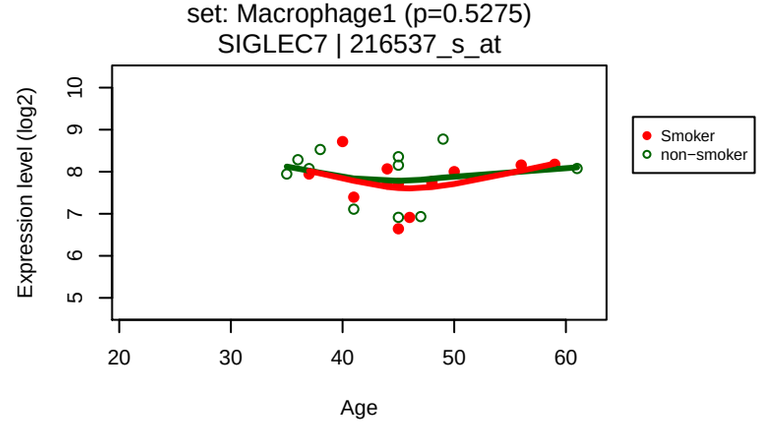

set: SmallAirway1 (p=0.2824)
SIGLEC7 | 216537_s_at
Expression level (log2)
Age
Smoker
non−smoker

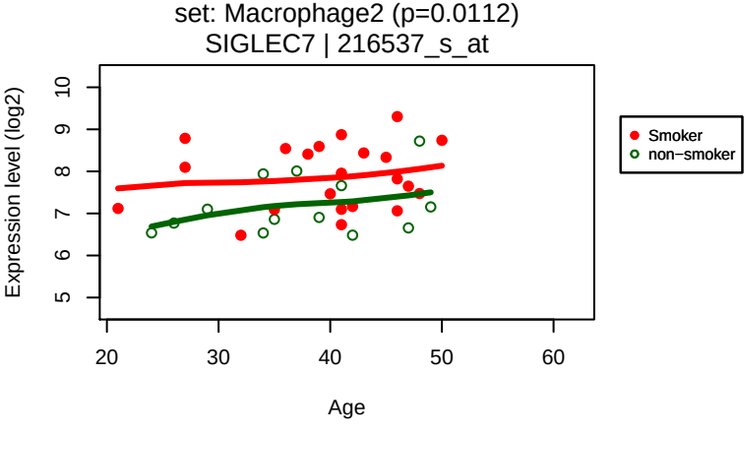

set: SmallAirway2 (p=0.3884)
SIGLEC7 | 216537_s_at
Expression level (log2)
Age
Smoker
non−smoker

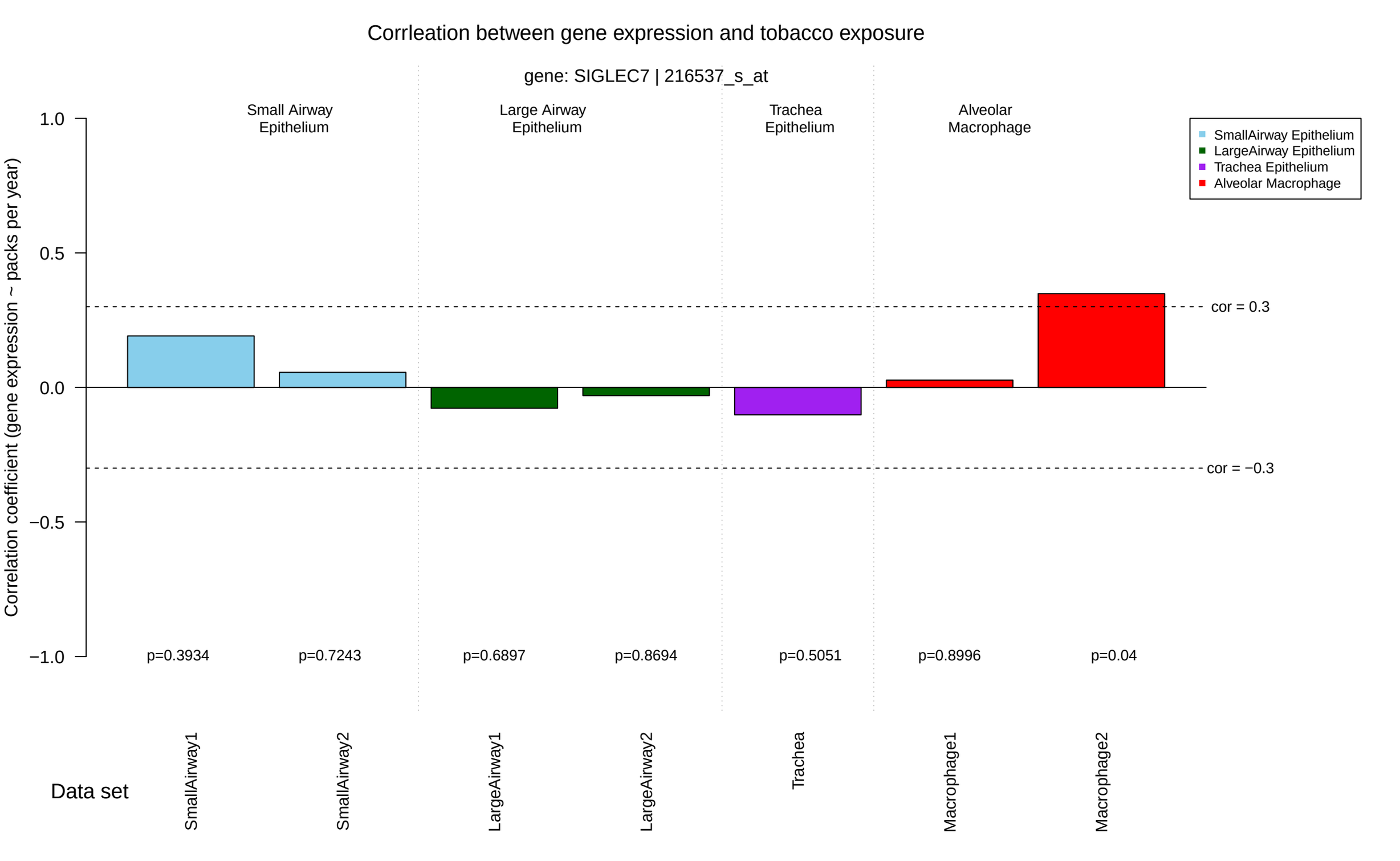

set: LargeAirway1 (p=0.0448)
SIGLEC7 | 216537_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4701)
SIGLEC7 | 216537_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4777)
SIGLEC7 | 216537_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5275)
SIGLEC7 | 216537_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0112)
SIGLEC7 | 216537_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIGLEC7 | 216537_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3934
p=0.7243
p=0.6897
p=0.8694
p=0.5051
p=0.8996
p=0.04
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2